```
> restart;#grating;
> 180· arcsin(0.532)/Pi;
```

32.14068662 **(1)**

```
> 180· arcsin(0.690)/Pi;
```

43.63010887 **(2)**

```
> 180· arcsin(0.450)/Pi;
```

26.74368394 **(3)**

```
> 180·arcsin(2·0.532)/Pi;
```

89.99999998 − 20.39097947 I **(4)**

```
> 180·arcsin(2·0.659)/Pi;
```

89.99999998 − 44.56152367 I **(5)**

```
> 180· arcsin(0.0015)/Pi;#delta less than θ =0.15 degrees thus ~2.2 nm
```

0.08594370151 **(6)**

```
> 180· arcsin(0.5342)/Pi;
```

32.28967364 **(7)**

```
> 180· arcsin(0.5364)/Pi;
```

32.43890587 **(8)**

```
> Eμ := evalf(1239/532 − 1239/534);#μ neutrino energy eV oscilation
```

$E\mu := 0.008722649320$ **(9)**

```
> 180· arcsin(0.536)/Pi − 180· arcsin(0.534)/Pi;
  # 0.135 degrees due to the μ neutrino oscilation
```

0.13563513 **(10)**

```
>
> 180· arcsin(0.532)/Pi − 180· arcsin(0.534)/Pi
```

−0.13543265 **(11)**

```
> Eμ := evalf(1239.84/532 - 1239.84/534);
```

$$E\mu := 0.008728563 \tag{12}$$

```
> Eτ_0 := evalf(1239.84/690);
```

$$E\tau_0 := 1.796869565 \tag{13}$$

```
> evalf(1.975·1.6295)^(1/2)
```

$$1.793951644 \tag{14}$$

```
> Eμ := evalf(Eτ_0/2);
```

$$E\mu := 0.8984347825 \tag{15}$$

```
> a := evalf(0.9/0.00895);
```

$$a := 100.5586592 \tag{16}$$

```
>
>
> solve(0.188/(1 - Vτ^2/c^2)^(1/2) = 1.975, Vτ);
```

$$-0.9954591343\ c,\ 0.9954591343\ c \tag{17}$$

```
> Vτ := 0.9954591343 ·299792458
```

$$V\tau := 2.984311407 \times 10^{8} \tag{18}$$

```
> E0 := 1239.84/690.69;
```

$$E0 := 1.795074491 \tag{19}$$

```
>
> (1.9774·1.6295)^0.5 ;# The geomeric mean between τ, μ, ν
```

$$1.795041309 \tag{20}$$

```
> #1239.84/1340.4;
> solve(0.188/(1 - V0τ^2/c^2)^(1/2) = 1.975);
```

(21)

$$\{V0\tau = -0.9954591343\ c,\ c = c\},\ \{V0\tau = 0.9954591343\ c,\ c = c\} \quad (21)$$

```
> V0tau := 0.995459134*299792458;
```

$$V0tau := 2.984311406 \times 10^{8} \quad (22)$$

```
> solve(0.00038/(1-Vg^2/c^2)^(1/2) = 1.6295, Vg);
```

$$-0.9999999728\ c,\ 0.9999999728\ c \quad (23)$$

```
> Vg := 0.9999999728*299792458;
```

$$Vg := 2.997924498 \times 10^{8} \quad (24)$$

```
> solve(0.0087/(1-Vμ^2/c^2)^(1/2) = 1.795, Vμ);
```

$$-0.9999882542\ c,\ 0.9999882542\ c \quad (25)$$

```
> Vμ := 0.9999882542*299792458;
```

$$V\mu := 2.997889367 \times 10^{8} \quad (26)$$

```
> 1239.84/3.59;#Eμ = 3.59eV ⇒ 345.35 nm
```

$$345.3593315 \quad (27)$$

```
>
> # a rough estimation of photon rest mass supposing the same ratio
    photon magnetron as Plank and magnetic constant
    in eV less than ~10^-9 eV 1/c^2
> h·ν = hm·100·Mhz;
    # considering enegy of 620nm equal to electromagntic energy of
    frequency of 100 Mhz
```

$$h\,\nu = 100\ hm\ Mhz \quad (28)$$

```
> 299792458/(620·10^-9·100·10^6·4794614.95);
```

$$1.008498550 \quad (29)$$

```
> 0.0087/4794614.95;#mphot555nm/mmag112Mhz rest mass ratio between photon
    and magnetron;
```

(30)

$$1.814535701 \times 10^{-9} \quad \textbf{(30)}$$

```
> 0.00678/4794614.95# the rest mass of π^0 is evan smaller
    ~0.00678 eV/c^2 thus photon rest mass is, a rough estimation,
    ~10^-9 eV/c^2;
```

$$1.414086443 \times 10^{-9} \quad \textbf{(31)}$$

```
> # The lowest photon energy where the photon interfered= 0.925eV
> h := 6.626070150 × 10^-34; c := 299792458; ec := 1.602176634 × 10^-19;
    h·c/(690·10^-9·ec);
```

$$h := 6.626070150 \times 10^{-34}$$

$$c := 299792458$$

$$ec := 1.602176634 \times 10^{-19}$$

$$1.796872441 \quad \textbf{(32)}$$